

Exact Ranks for the Dyadic Sections of Euler's Totient

An unconditional infinite-dimensionality theorem, an explicit denominator bound, and the weight that separates Erdős #249 from its solved neighbours

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ABSTRACT

Erdős #249 asks whether $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational. It is open, and nothing here closes it. What is established, unconditionally and with machine-checked proofs, is the following. For every $e \geq 0$ the canonical level- e family of dyadic totient sections $n \mapsto \varphi(2^j n + r)$ is linearly independent over $\mathbb{Q}$ with span of *exact* dimension $2^e + 1$; consequently the span of all dyadic sections of φ is not finite-dimensional over $\mathbb{Q}$. Separately, $S \neq p/q$ for every $q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$.

We also record the exact identity $S = \sum_{d \geq 1} A(d)/(2^d - 1)$ with $A = \varphi * \mu \geq 0$, $A(p) = p - 2$, $A(p^e) = (p - 1)^2 p^{e-2}$. This places #249 in one family with three series that *are* settled — the Erdős–Borwein constant, and, since December 2025, the ω and Ω series of Tao and Teräväinen — and isolates one difference: those three have a 0/1 Lambert weight, and A is unbounded. For *eventually periodic* nonnegative weights the irrationality conclusion is itself machine-checked here in every integer base $b \geq 2$; A is not eventually periodic, and that gap is the problem.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

1 The four series, and which are settled

If $f = g * \mathbf{1}$, that is $f(n) = \sum_{d|n} g(d)$, then interchanging the two sums gives

$$\sum_{n \geq 1} \frac{f(n)}{2^n} = \sum_{d \geq 1} \frac{g(d)}{2^d - 1}. \quad (1)$$

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Four classical coefficient sequences are of this shape. The table records the resulting constant, the weight $g = f * \mu$ it corresponds to, and its status. Every decimal below was recomputed to 50 places for this note.

f	$\sum f(n)/2^n$	weight $g = f * \mu$	status
τ	$1.60669515 \dots = E$	$g \equiv 1$	Irrational (Erdős 1948); formalised here
ω	$0.51694281 \dots$	$g = \mathbf{1}_{\text{primes}}$	Irrational (Tao–Teräväinen 2025) [1]
Ω	$0.58950329 \dots$	$g = \mathbf{1}_{\text{prime powers}}$	Irrational (<i>ibid.</i> , remark)
φ	$1.36763080 \dots = S$	$g = A = \varphi * \mu$	Open (Erdős #249)

The first three weights take only the values 0 and 1. The fourth does not: $A(p) = p - 2$ is unbounded. That is the only structural difference between #249 and its three settled neighbours, and it is not a cosmetic one — the settled cases are all instances of Erdős #257, the question of which *supports* A make $\sum_{a \in A} (2^a - 1)^{-1}$ irrational, whereas #249 is a single unbounded-weight member of the same family.

Proposition 1.1 (the #249 weight). $S = \sum_{d \geq 1} A(d)/(2^d - 1)$, where $A = \varphi * \mu$ satisfies $A \geq 0$, $A(p) = p - 2$ and $A(p^e) = (p - 1)^2 p^{e-2}$ for $e \geq 2$.

This is (1) at $f = \varphi$, using $\varphi = A * \mathbf{1}$; it is checked as the [positive-weight lift](#). Two further rungs of the same ladder are exact rationals: $\sum_{d \geq 1} \mu(d)/(2^d - 1) = \frac{1}{2}$ ([Lean](#)) and $\sum_{d \geq 1} \varphi(d)/(2^d - 1) = 2$ ([Lean](#)), and there is a signed squared-Mersenne lens $S = \frac{1}{2} + \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$ ([Lean](#)). The difficulty of #249 therefore does not live in the Lambert transform, where the neighbouring values are rational and exactly computable; it lives on the power-series side.

Why the unboundedness is the obstruction, precisely. For nonnegative rational weights that are *eventually periodic*, and with a positive value at a positive index of the periodic part, irrationality of $\sum_a g(a)/(b^a - 1)$ is proved here for every integer base $b \geq 2$ ([Lean](#); Luca and Tachiya prove a broader periodic rational-coefficient theorem by different methods [3], and no priority is claimed). So the weighted problem is settled on the eventually periodic class and open on A . What fails for A is not positivity and not rationality of the values: it is that no bounded amount of the weight determines the rest. Section 2 makes that last sentence a theorem.

2 The rank theorem

For $j \geq 0$ and $0 \leq r < 2^j$ write $\varphi_{j,r}$ for the dyadic section $n \mapsto \varphi(2^j n + r)$, regarded as an element of the $\mathbb{Q}$ -vector space of sequences $\mathbb{N} \rightarrow \mathbb{Q}$.

Theorem 2.1 (exact level rank). *For every $e \geq 0$ the canonical level- e family of dyadic totient sections is linearly independent over $\mathbb{Q}$, and the dimension of its span is exactly $2^e + 1$.*

Corollary 2.2 (infinite dimensionality). *The span of $\{\varphi_{j,r} : j \geq 0, 0 \leq r < 2^j\}$ is not finite-dimensional over $\mathbb{Q}$.*

The independence is [checked](#), the exact dimension is [checked](#), and the corollary is [checked](#). The corollary follows from the theorem in one step: were the full span finite-dimensional of dimension d , the level- d family would embed in it and force $2^d + 1 \leq d$.

The mechanism is a separated evaluation minor. For each level one produces, by the Chinese remainder theorem together with Dirichlet's theorem on primes in arithmetic progressions, a set of evaluation points at which the section matrix has a nonzero minor with the required separation ([Lean](#)). Even residues reduce to odd-core channels, so the count $2^e + 1$ rather than 2^{e+1} is exact rather than an estimate.

Remark. Theorem 2.1 is a statement about φ , not about S : it is true, and its proof is complete, whether or not S is irrational. It is included here because it is the sharpest available form of “no finite amount of the totient's dyadic structure determines the rest”, which is the property the eventually periodic weight theorem consumes and A lacks.

The rationality-side consequence. Binary long division turns a hypothetical rational value into an integer carry recurrence. Transporting Theorem 2.1 through that recurrence gives: if S were rational, one positive-multiplier tempered integral carry would have canonical carry-section span of dimension at least $2^e - 1$ for every e ([Lean](#), using the [rank transport](#)). This is an anti-compression barrier, and we state its limit plainly: it rules out every rationality proof that would exhibit a carry of *bounded* rank, and it is not an irrationality criterion, because no finite-rank upper bound for every rationality-supplied carry is proved.

3 The denominator bound, and what it is worth

Theorem 3.1 (denominator exclusion). *If $S = p/q$ with $p \in \mathbb{Z}$ and $q \geq 1$, then*

$$q > 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

This is [checked](#) as a single statement about S .

We state its provenance rather than leaving a reader to find it. The proof is an exact Farey-window exclusion: the underlying Lean lemma is a gap check over the Farey window $[1, 240]$, and the argument is the elementary one — a rational with small denominator cannot lie in a sufficiently narrow interval around a well-approximated real, and enough digits of S are computable to make the interval narrow. *The method is standard, and we claim no novelty for it.* What the entry contributes is the explicit constant at this scale, discharged inside a kernel-checked proof rather than by a script. A reader who regards an explicit Farey exclusion as routine is not in disagreement with this note; Theorem 2.1, not Theorem 3.1, is where we would ask a sceptical reader to look.

The bound transfers verbatim to the probabilistic form of the constant: with X, Y independent and $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ one has $S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1)$, so that probability has no rational representation of denominator at most 39 819 823 323 350 687 661 677 887 437 915 526.

4 Two no-go theorems about methods

Two negative results are unconditional theorems, and they are the kind of statement whose scope is easy to overstate, so each is given with the exact class it closes.

Proposition 4.1 (fixed-precision transport). *At every fixed positive precision, each finite compatible valuation-unit word admits a prefix-locked centred completion.*

(Lean.) Closes: arguments that would derive a contradiction from a local signature read at a precision fixed in advance. Does *not* close: arguments whose precision grows with the scale, or which use global arithmetic information about φ .

Proposition 4.2 (factor ideals and finite-rank shift algebra). *For every $t \geq 3$ there is a nonzero synthetic affine carry whose forcing letters lie in the principal ideal generated by $\varphi(\text{lcm}(1, \dots, t))$, which matches every stated whole-ray anchor, stays within the diagonal bounds, and whose state is a strict survivor; the same countermodel survives every finite integer shift polynomial.*

(Lean.) Closes: factor-ideal, whole-ray-anchor and finite-rank shift-algebra arguments taken alone. Does *not* close: anything nonlinear in fresh divisor channels. The construction is synthetic — it does not assert that its forcing letters occur as actual totient differences — and that limitation is part of the statement, not a caveat attached to it.

5 What remains open

The problem is open and this note does not narrow it. Precisely:

1. Irrationality of S . No proof is claimed, and Theorem 3.1 bounds a denominator without excluding one.
2. The certificate route reduces irrationality to an exact condition: a finite integer separation calculation, with a rigorous tail bound, occurring beyond every cutoff. Finite instances are checked; the unbounded supply is not, and *an equivalence is not progress*. We flag this explicitly because the reduction is the part of this work most likely to be mistaken for a result.
3. The natural analytic target — a cancellation estimate for the first-harmonic block sum attached to the doubling orbit of $(2^h - 1)S$ — is unproved, and is strictly stronger than irrationality rather than equivalent to it.

A route we would now take seriously, and do not have: adapt the Gowers-uniformity and quantitative-correlation method of [1] from the 0/1 weight $\mathbf{1}_{\text{primes}}$ to an unbounded weight. Their argument uses the prime-dilation identity $\omega(pn) = \omega(n) + 1 - \mathbf{1}_{p|n}$, whose main increment is constant and whose error is sparse. The corresponding statement for φ has a valuation-dependent defect and no such clean form, so the transfer is not routine; but it is the first place a reader who wants to close #249 should look, and it did not exist when this development began.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions, and kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. Numerical decimals quoted in Section 1 were recomputed independently for this note. Erdős Problem #249 remains open.

References

- [1] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Proves irrationality of $\sum_{n \geq 1} \omega(n)/2^n$, and remarks that the method also gives $\sum_{n \geq 1} \Omega(n)/2^n$ and every integer base $b \geq 2$.
- [2] K. Pratt, *The irrationality of a prime factor series under a prime tuples conjecture*, arXiv:2409.15185. The conditional predecessor of the above.
- [3] F. Luca and Y. Tachiya, *Irrationality of Lambert series associated with a periodic sequence*.
- [4] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 61.
- [5] T. Bloom, *Erdős Problems*, <https://www.erdosproblems.com/249>.